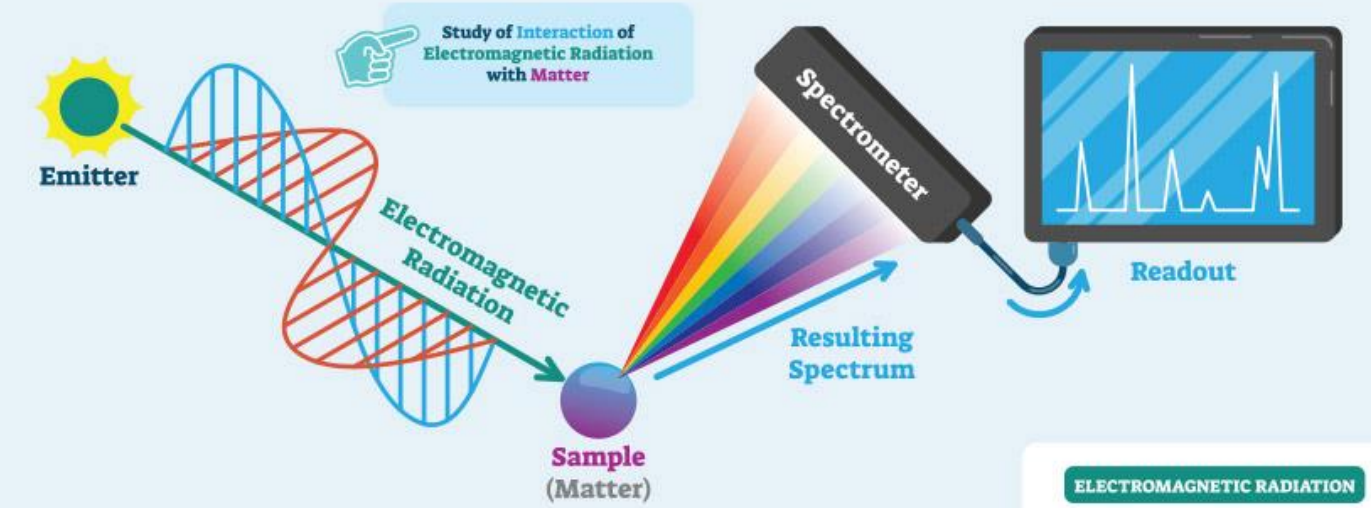
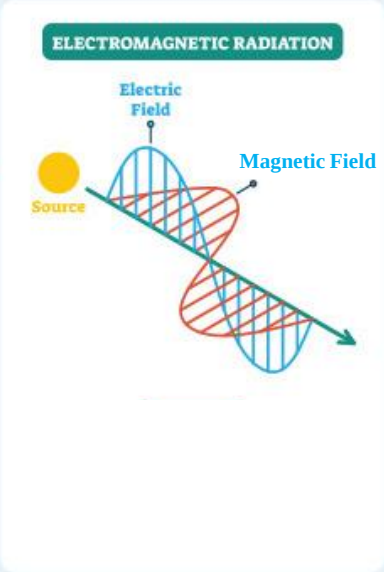
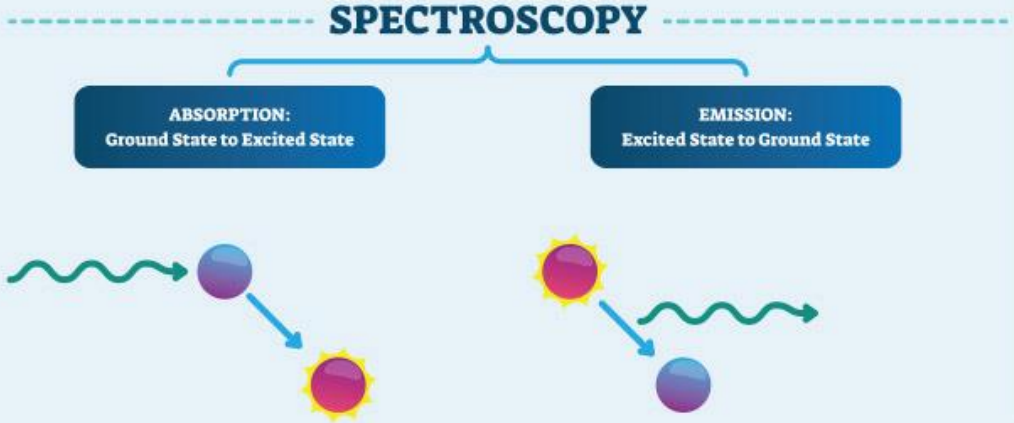
Fundamentals of Spectroscopy
CH2201

1. Introduction
2. Atomic Structure and Atomic Spectra
3. Rotational Spectroscopy
4. Vibrational spectroscopy
5. Raman Spectroscopy
6. Electronic spectroscopy

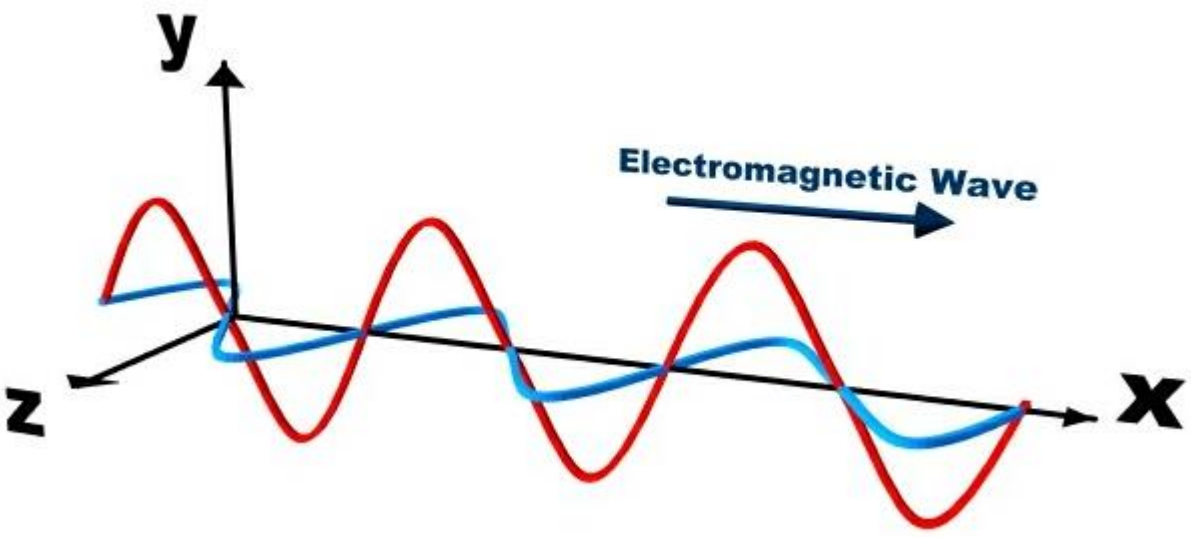
WHAT IS **SPECTROSCOPY?**



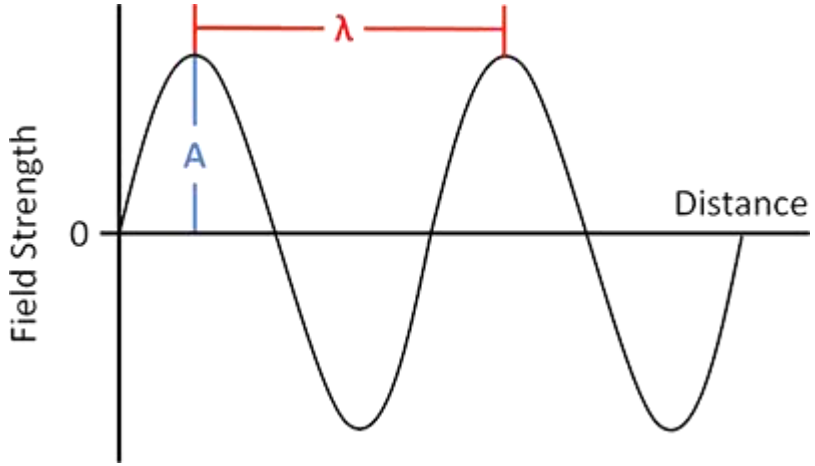




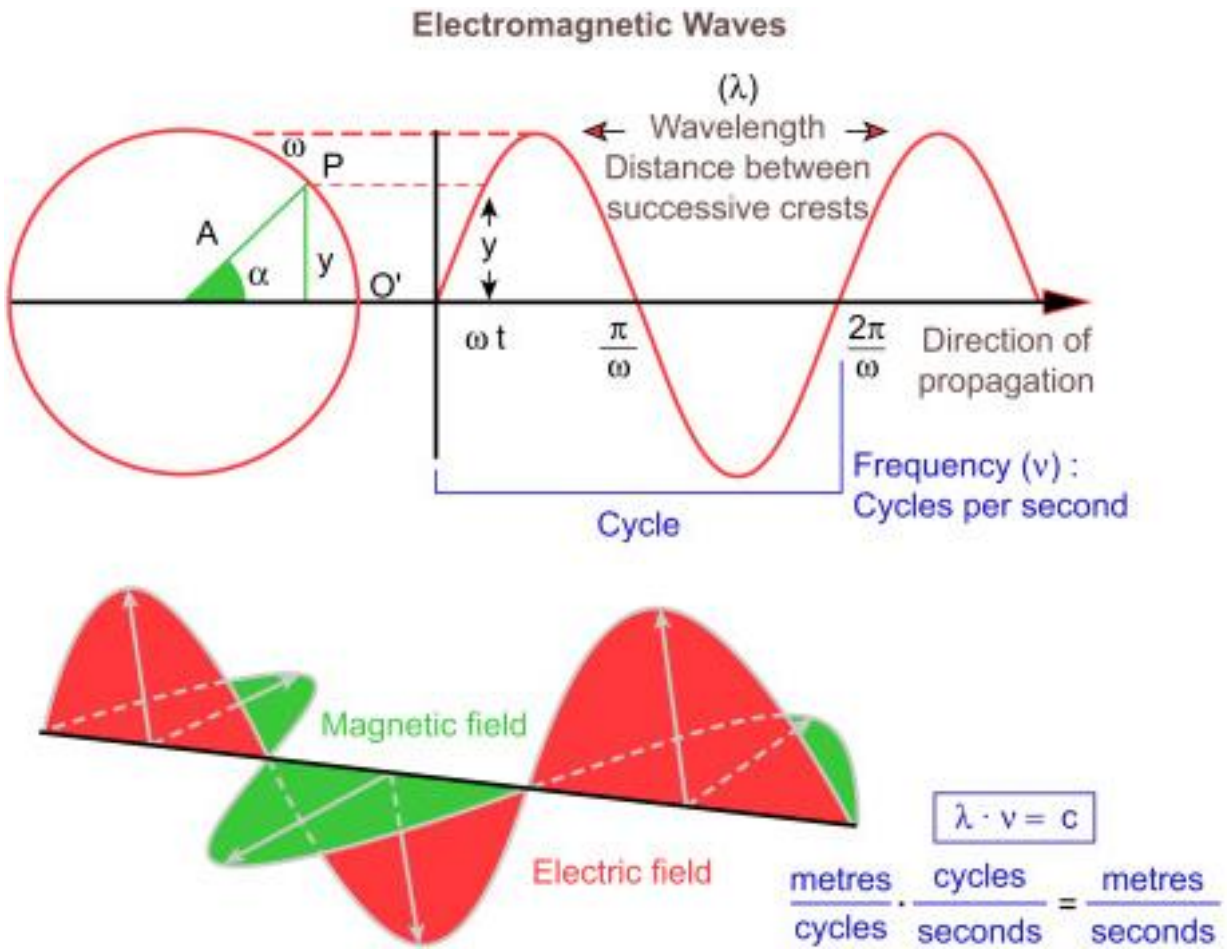
Electromagnetic Radiation



Electromagnetic radiation is a **simple harmonic wave** propagated from a source and travelling in a straight line when reflected or refracted.



Simple Harmonic Wave has properties of the sin wave, defined by, $y = Asin\theta$.
y is the displacement with a maximum value A, and θ is an angle varying between 0 and 360°.



A point **P** travels with a uniform angular velocity ω rad s^{-1} in a circular path of radius A.

The time is measured when **P** passes O' and after t sec, i.e., **P** has described an angle $\theta = \omega t$ radians.

The vertical displacement of **P** is

$$y = A sin\theta = A sin\omega t$$

After a time of $2\pi/\omega$ sec, **P** will return to O' completing a cycle.

In 1 sec the pattern will repeat itself $\omega/2\pi$ times, referred to as the frequency (ν) of the wave.

Hence, $\boxed{y = A sin\omega t = A sin 2\pi\nu t}$ ⟹ **Basic equation of wave motion.**

Nature of a Travelling Wave

The **distance variation of displacement with time** is given by the relation:

$$x = ct$$

x is the distance covered in time t at a speed c.

Hence,

$$y = Asin2\pi\nu t = Asin\frac{2\pi\nu x}{c}$$

When the velocity is c m/sec and there are ν cycles/sec, there are evidently ν waves in c metres.

$\nu\lambda = c$ or $\lambda = {}^{c}/_{\nu}$ metres

$$\boxed{y = Asin\frac{2\pi x}{\lambda}}$$

and in terms of **wavenumber ($\bar{\nu}$)**

$\bar{\nu} = 1/\lambda$ cm^{-1}

$$\boxed{y = Asin2\pi\bar{\nu}x}$$

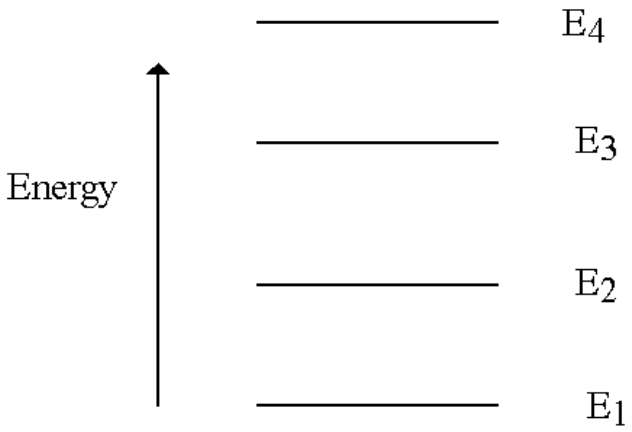
Energy and Quantization

Energy in the form of electromagnetic radiations comes in **discrete units called photons**.

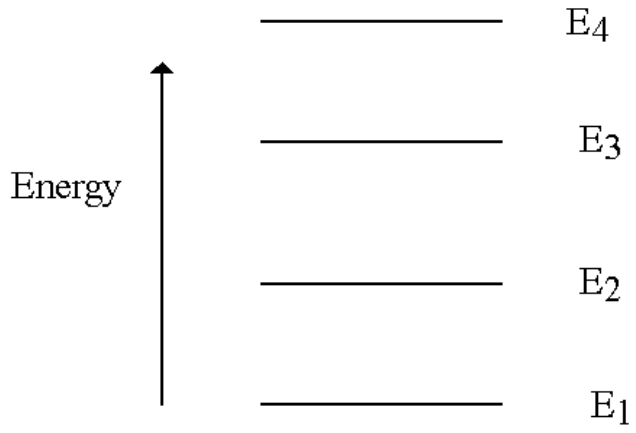
Each **photon** contains an amount of energy called **quanta** determined by the frequency of electromagnetic radiation.

The **equation relating to frequency and energy** is given by: $\boxed{E = h\nu = h\frac{c}{\lambda} = hc\bar{\nu}}$ (Max Planck)

The **energy of an oscillator is discontinuous** and any **change in its energy content** can occur only by means of a **jump between two distinct energy states**.

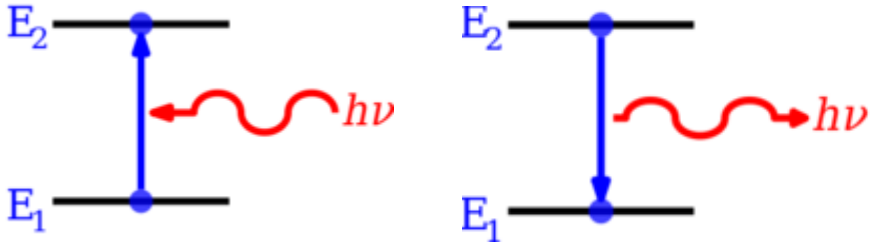


Consider possible **discrete energy states** of a system where the suffixes represent the **quantum numbers**.



Transitions can take place **between the levels** provided the appropriate amount of energy, $\boldsymbol{\Delta E = E_2 - E_1}$, can be **either absorbed or emitted by the system**.

Planck suggested that such absorbed or emitted energy can take the form of electromagnetic radiation and the **frequency of radiation will be:** $\boldsymbol{\nu = \Delta E / h}$ **Hz, i.e.,** $\boldsymbol{\Delta E = h\nu}$ **joules**



h is the Planck's constant = 6.63 x 10^{-34} joules s $molecule^{-1}$

Born-Oppenheimer Approximation

….is based on the **comparative masses of nucleus and electrons**

- Nuclei are much more massive than electrons (~2,000:1 or more)
- Electron motions much faster than nuclear motions
- To a first approximation, assume electrons move "instantly" compared to nuclei. Equivalently, nuclei frozen compared to electrons

Max Born

J. Robert Oppenheimer

The total energy (E) of the system is thus the **sum** of energies due to **electronic (E_{elec}) and nuclear (E_{nuc})** motions.

$$E = E_{elec} + E_{nuc}$$

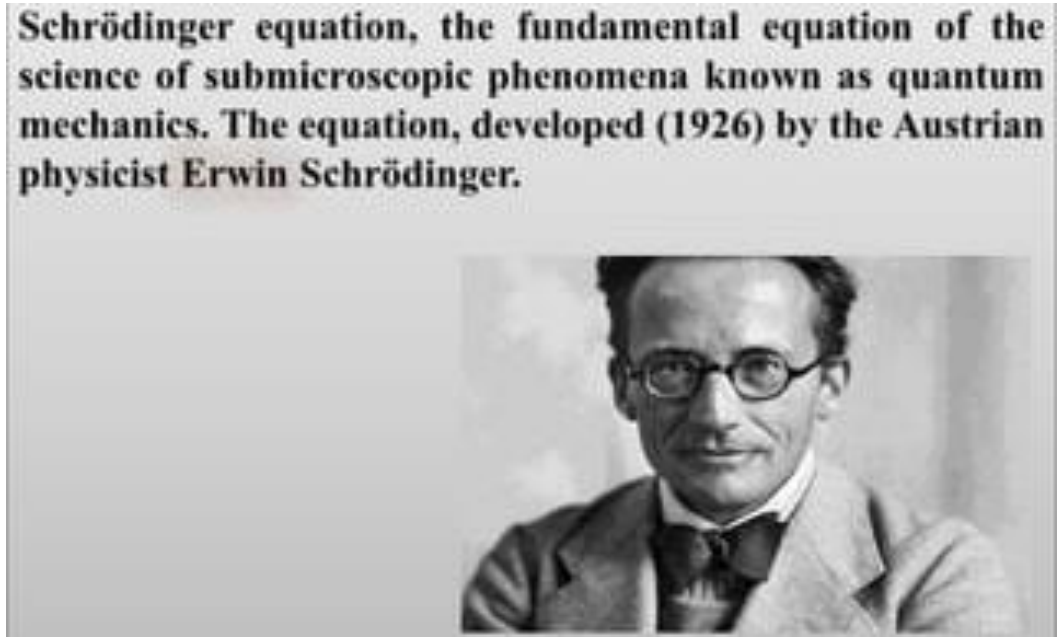
where, $E_{nuc} = E_{trans} + E_{vib} + E_{rot} + E_{introt}$

where, E_{trans}, E_{vib}, E_{rot} and E_{introt} are for translational, vibrational, rotational and internal rotational energies, respectively

E_{trans} is omitted since the gap between successive translational energy levels is so small that it can be considered as continuous (not quantized).

Schrödinger Wave Equation

….is a **second degree differential equation** and the **most appropriate method** of providing the **solution of one body problems**



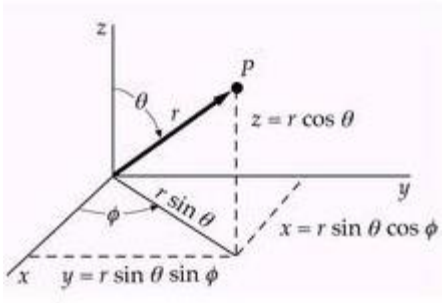
The time independent form of wave equation in **one** and **three dimensions** is given by

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2 m}{h^2}(E-U)\psi = 0 \quad \text{and} \quad \nabla^2\psi + \frac{8\pi^2 m}{h^2}(E-U)\psi = 0$$

ψ is the wave function which describes the wave property of a particle with mass m, ∇^2 is the Laplacian operator expressed as

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \qquad Cartesian\ coordinates$$

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2 sin\theta}\frac{\partial}{\partial\theta}\left(sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2 sin^2\theta}\frac{\partial^2}{\partial\varphi^2} \qquad Polar\ coordinates$$



where, $x = rsin\theta cos\varphi$, $y = rsin\theta sin\varphi$, $z = rcos\theta$; $d\tau = r^2 sin\theta dr d\theta d\varphi$

$$\nabla^2\psi + \frac{8\pi^2 m}{h^2}(E - U)\psi = 0$$

Rearranging this equation, we get

$$\left(\frac{h^2}{8\pi^2 m}\nabla^2 + U\right)\psi = E\psi$$ Or $$\hat{H}\psi = E\psi$$ where, $$\hat{H} = \frac{h^2}{8\pi^2 m}\nabla^2 + U$$ (Hamiltonian operator)

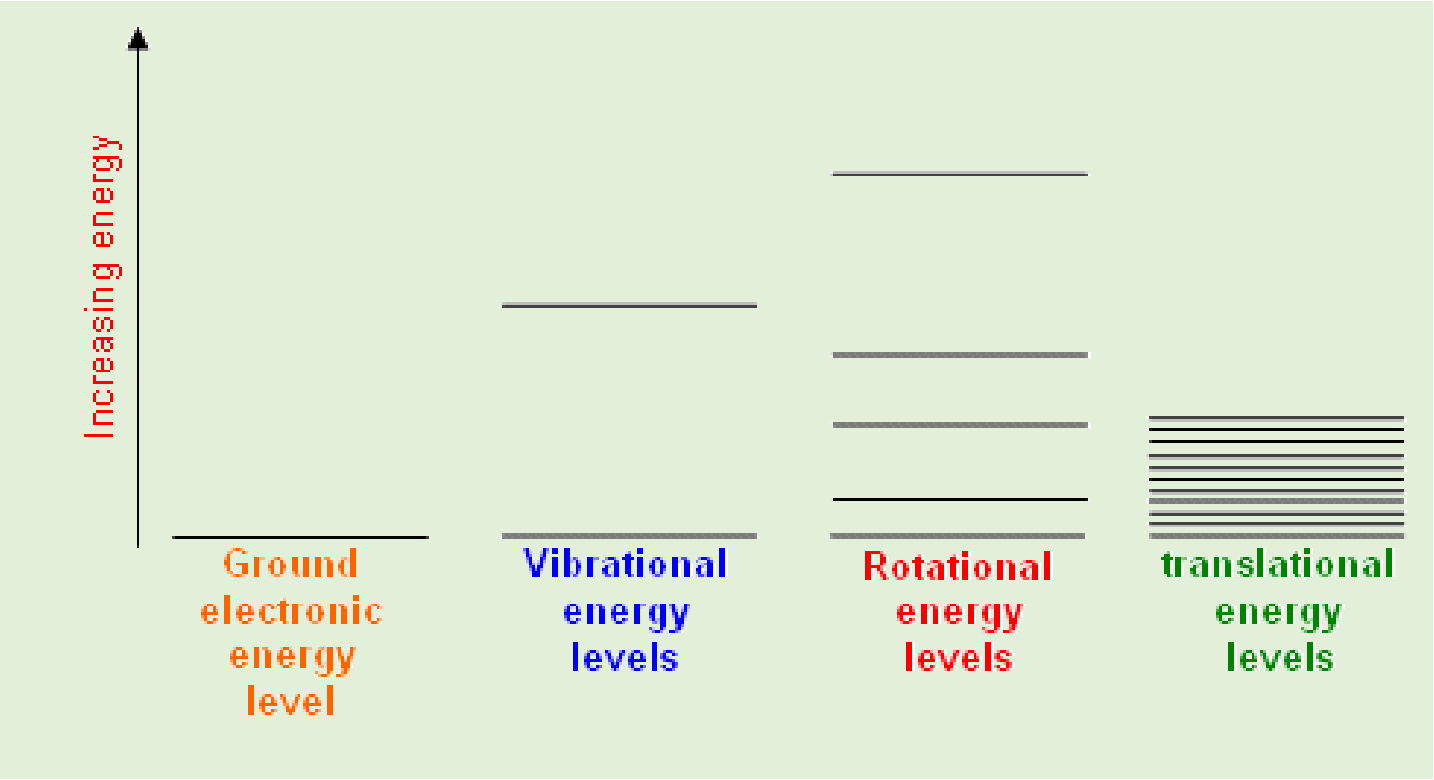
The **wavefunction ψ** is associated with **a state of physical system corresponding to energy *E***, i.e., it represents the **energy state of the system**.

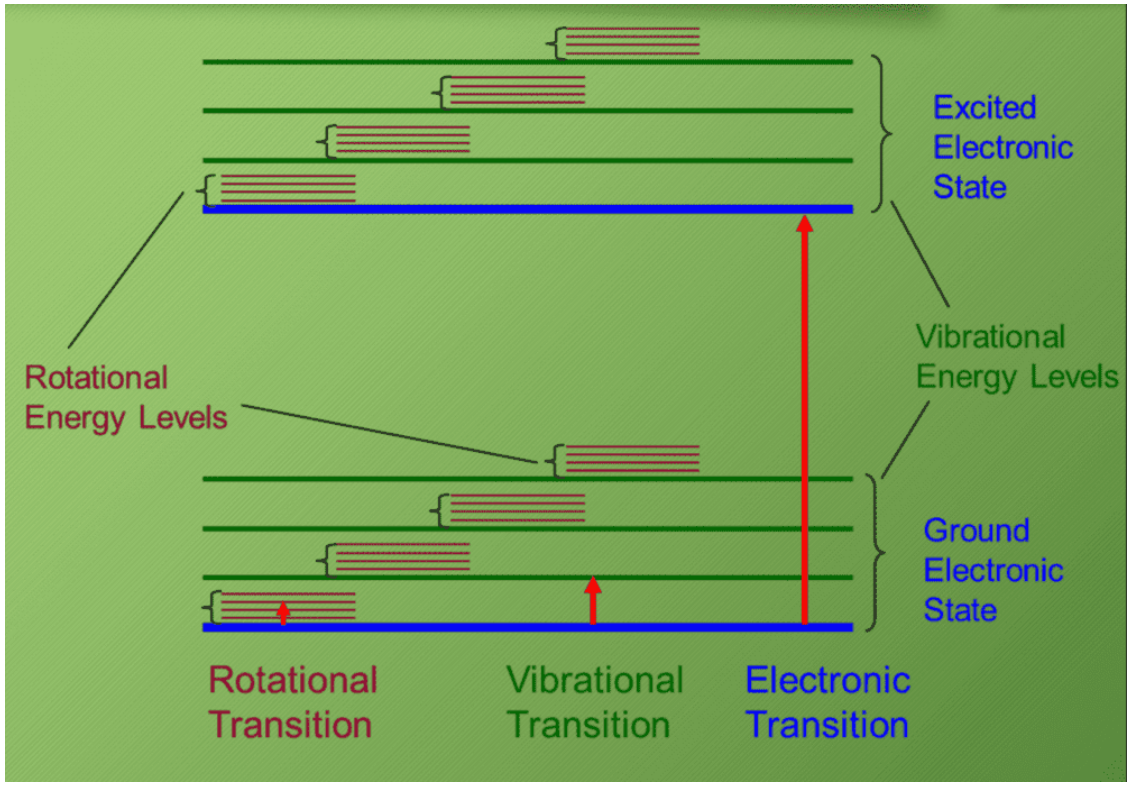
The energy of various types of motions, viz., electronic, vibrational, rotational, etc., can be determined with the help of Schrödinger wave equation.

It is found that in case of molecules:

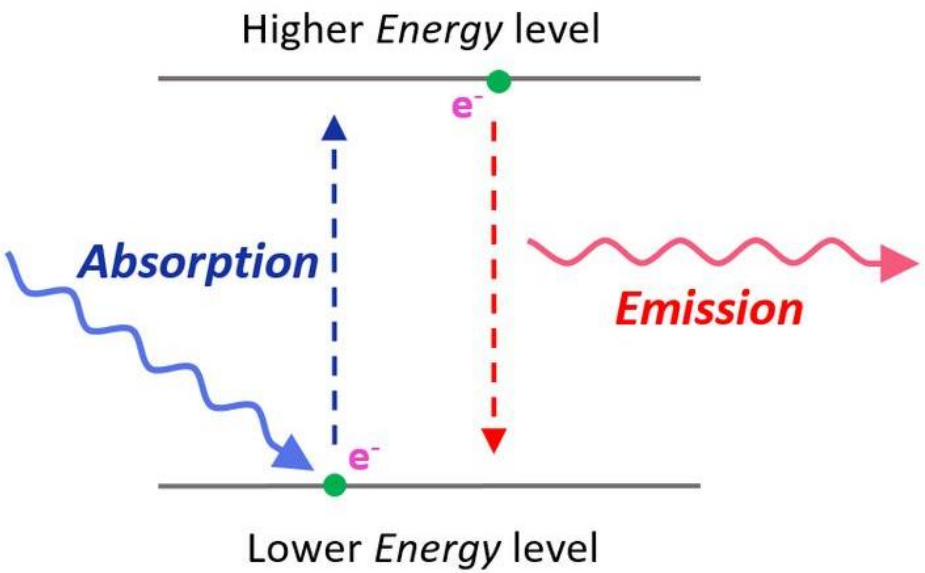
$$\boxed{E_{elec} > E_{vib} > E_{rot}}$$

$E_{elec} > E_{vib} > E_{rot}$





General Condition of Resonance: Absorption and Emission

When any atomic or molecular system is exposed to electromagnetic radiation the **absorption or emission of a photon with frequency (ν)** will occur only if the **energy difference (ΔE) between the ground and the excited states of the system matches the energy of the photon**.

$$\pm\Delta E = E_2 - E_1 = h\nu \quad \text{or} \quad \Delta E = |E_1 - E_2| = h\nu$$